Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Numerical Methods

Medium & Hard • 25 Questions

Scan for free resources

Section A – Medium (Q1–15)

1. Show $x^3 - 3x - 1 = 0$ has a root in $[1, 2]$. **[3]**

2. Show $x^3 + x - 1 = 0$ has a root in $[0, 1]$. **[3]**

3. Show $e^{-x} = x$ has a root in $[0.5, 0.6]$. **[3]**

4. Show $\cos x = x$ has a root in $[0.7, 0.8]$. **[3]**

5. Iteration $x_{n+1} = \frac{1}{2}(x_n + \frac{7}{x_n})$, $x_0 = 3$. Find x_1, x_2, x_3 to 4 d.p. **[5]**

6. Iteration $x_{n+1} = \sqrt{2 + x_n}$, $x_0 = 1$. Find x_1, x_2, x_3 to 3 d.p. **[4]**

7. Iteration $x_{n+1} = \frac{x_n^2 + 2}{2x_n}$, $x_0 = 1.5$. Find x_1, x_2 to 4 d.p. **[4]**

8. Trapezium rule, 4 strips: $\int_0^2 \sqrt{1 + x^3}\,\mathrm{d}x$ to 3 d.p. **[5]**

9. Trapezium rule, 4 strips: $\int_0^1 e^{-x^2}\,\mathrm{d}x$ to 3 d.p. **[5]**

10. Trapezium rule, 5 strips: $\int_1^2 \frac{1}{x}\,\mathrm{d}x$ to 3 d.p. **[5]**

11. Newton-Raphson, $x_0 = 1.5$, root of $x^3 - 3x - 1 = 0$ to 3 d.p. **[6]**

12. Newton-Raphson, $x_0 = 1$, root of $x^3 - x - 1 = 0$ to 3 d.p. **[6]**

13. Newton-Raphson, $x_0 = 0.7$, root of $x = \cos x$ to 3 d.p. **[6]**

14. Show the equation $x^2 - 2 = 0$ has a root in $[1, 2]$ and use Newton-Raphson with $x_0 = 1.5$ to find it to 3 d.p. **[6]**

15. Use iteration $x_{n+1} = e^{-x_n}$ starting at 0.5 to find the root of $e^{-x} = x$ to 3 d.p. **[5]**

Section B – Hard (Q16–25)

16. Trapezium rule with 6 strips for $\int_0^{\pi} \sin x \, \mathrm{d}x$. Compare with exact value. **[6]**

17. Newton-Raphson for $x^3 - 2x - 5 = 0$, $x_0 = 2$, to 4 d.p. **[6]**

18. Show $f(x) = x^3 - x - 1$ has exactly one real root and find it to 3 d.p. using Newton-Raphson. **[7]**

19. Use the iterative formula $x_{n+1} = \sqrt[3]{x_n + 1}$ with $x_0 = 1$ to find a root of $x^3 - x - 1 = 0$ to 3 d.p. **[5]**

20. Trapezium rule, 4 strips: $\int_0^2 \frac{1}{1+x^2} \, \mathrm{d}x$. Compare with exact $\arctan 2$. **[6]**

21. Explain why the iteration $x_{n+1} = \cos x_n$ converges for the root of $x = \cos x$. **[4]**

22. Use Newton-Raphson to solve $e^x - 3x = 0$ starting at $x_0 = 1$, to 3 d.p. **[6]**

23. Find an interval of width 0.1 containing a root of $x^3 + x^2 - 1 = 0$. **[4]**

24. Use the formula $x_{n+1} = \frac{2x_n + 3/x_n^2}{3}$ (Newton for cube roots) with $x_0 = 1.5$ to approximate $\sqrt[3]{3}$ to 3 d.p. **[5]**

25. Trapezium rule with $n = 4$ for $\int_0^1 \sqrt{1+x}\,\mathrm{d}x$. Give answer to 3 d.p. **[5]**

End of Worksheet • *Show all working clearly.*